University of Glasgow

James Watt School of Engineering

ENG4053

Digital Signal Processing 4

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)	Date of Birth (DD/MM/YYYY)
2839768	09/03/2004

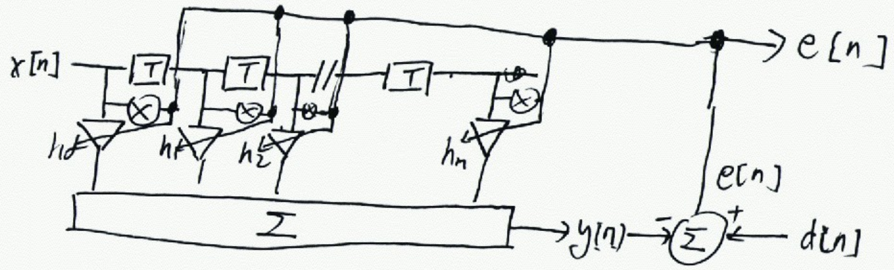
Mark with an **[X]** which question you have completed on this sheet. Cross **one box only**.

☐ Q1 ☐ Q2 ☒ Q3 ☐ Q4 ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE

You may use both sides but only answer **ONE** question per sheet

Q3.

(a). The dataflow diagram of an adaptive FIR/LMS filter:



The coefficients is changed according to:

$$\Delta h_m = -\mu \frac{\partial(\frac{1}{2}e(n)^2)}{\partial h_m} = \mu e(n) x(m-n)$$

To use this filter to remove background noise:

$x(n)$ = noise signal.

$d(n)$ = Sound recording + noise signal.

$e(n)$ = output signal.

In the process of updating the coefficients h_m, the $e(n) \to 0$. However, $e(n)$ will not exactly be zero, since any component that not correlate between $d(n)$ and $y(n)$ will be maintained, which is the sound recording. Therefore, the $e(n)$ is the cleaned up signal.

Sound rec. What is it? vocal?

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proper example!

(b). If the ~~the~~ learning rate is too low:

~~The filter, will~~ the response will converged slowly.

If the learning rate is too high:

the response may not be converged and could also be unstable.

If the reference is not limited to a max value, in the process of updating h_m, the coefficients and some other signal may overflow. This could lead to diverge.

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(c) The noise such as ~~constant~~ DC component and 50Hz powerline noise can be removed, since their frequency is known.

Other noise with the frequency that cannot be known before experiment or a wideband noise with similar shape as the wanted signal cannot be removed. This is because we need to provide a reference noise signal $x(n)$ to the LMS filter.

LMS = time-dom.

1/3

OB

corr

TOTAL 18

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Q Continued

(d) At initial stage:

$d(n) = \text{ECG signal} + 10mV \text{ noise}$

$x(n) = 1V$.

After converage:

$y(n) = 10\ mV$

so that the output signal:

$e_{ss}(n) = d(n) - y_{ss}(n) = \text{ECG signal}$.

Therefore, the coefficients should be like:

$h_m = \frac{y_{ss}(n)}{N x(n)} = \frac{10m}{100 \times 1} = 10^{-4}$

with length of 100.

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